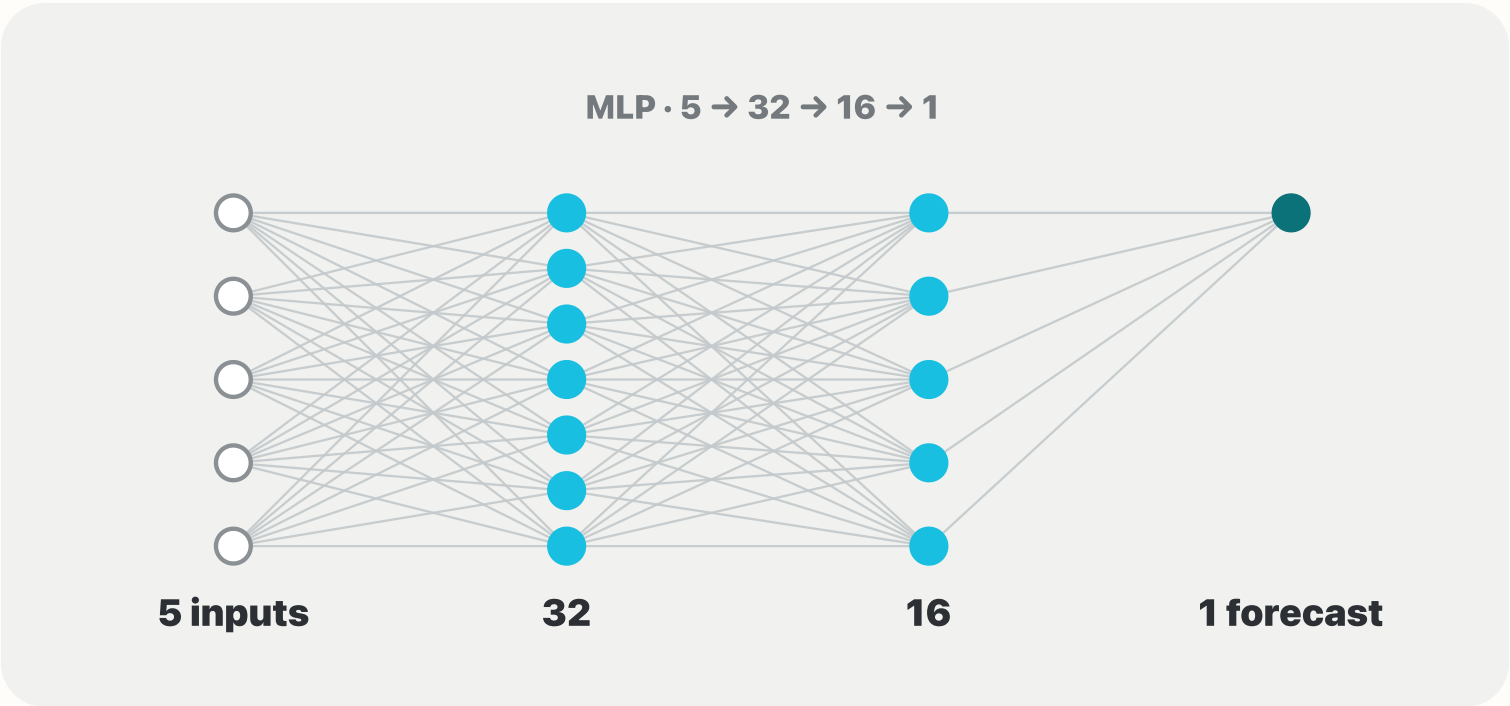


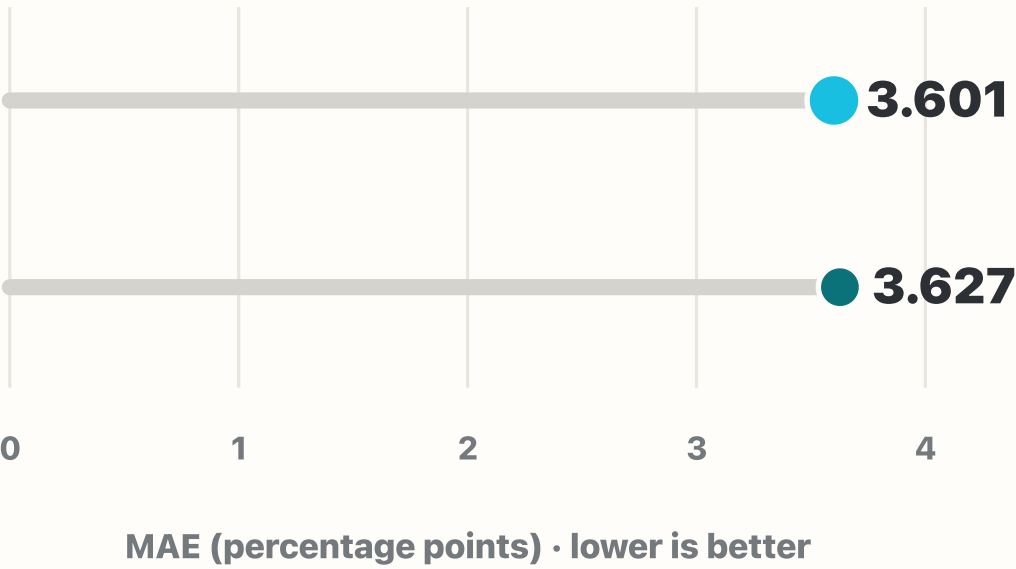
NEURAL NETS LEARN COMPLEX FUNCTIONS. MINE LOST MAE TO A CONSTANT.

I used quarterly bank reports to predict next-quarter deposit growth.



0% constant

MLP



THE CONSTANT PREDICTED 0% GROWTH FOR EVERY BANK.

DEPOSITS CAN LEAVE BEFORE LOANS PAY BACK.

Imagine the internet is down because the AI models broke it 🙄. You walk into your bank and ask for **\$15 in cash**.

YOU: "\$15 CASH, PLEASE." 



CASH IN THE VAULT

\$10



MONEY ALREADY OUT AS LOANS

\$90

\$15 withdrawal. \$10 cash.

**COOL. LET ME JUST TURN PART OF YOUR MORTGAGE INTO
A \$5 BILL.**

SO WHO KNOWS WHAT'S ON THE BALANCE SHEET?



Federal banking regulators collect standardized
Call Reports every quarter.

That gives us bank-by-bank snapshots of cash,
deposits, loans, capital, and more.

CASH

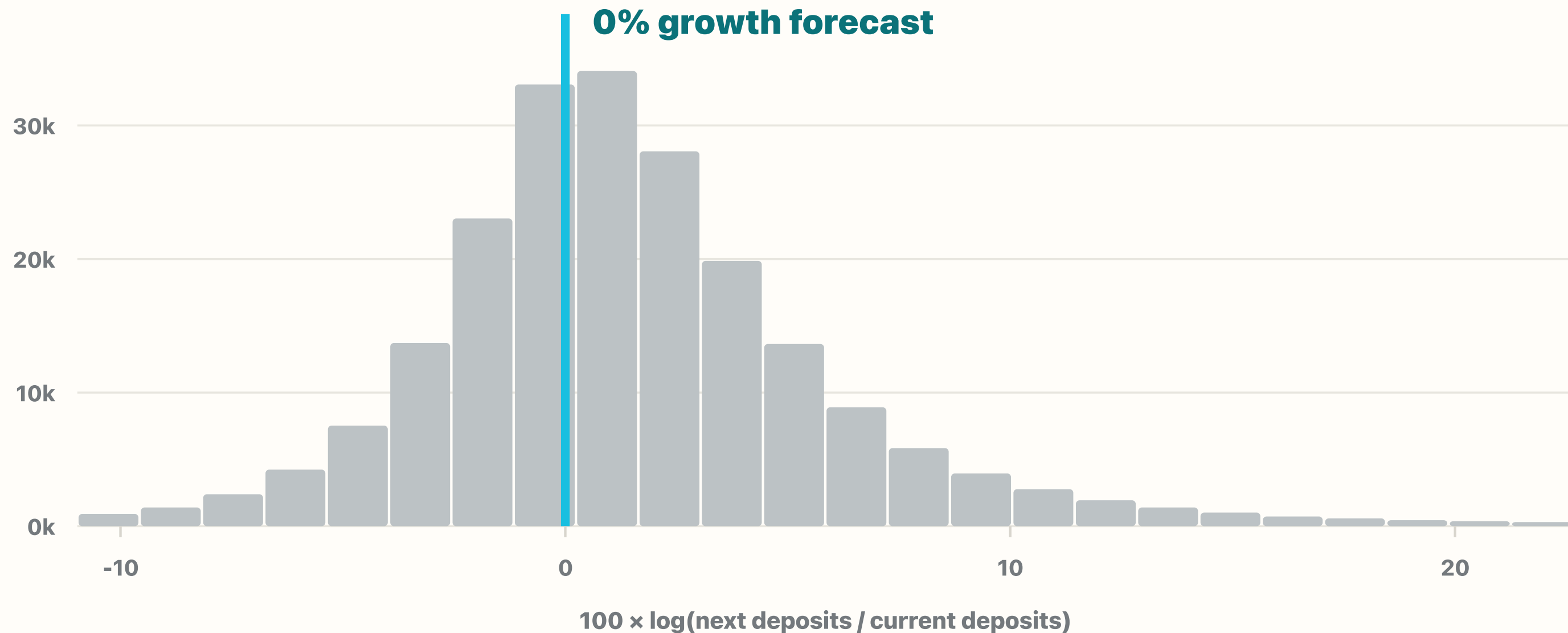
DEPOSITS

LOANS

CAPITAL

THAT'S WHY I CARE ABOUT WHERE DEPOSITS MOVE NEXT.

SO WHY WAS 0% GROWTH SO HARD TO BEAT?



A LOT OF THE TRAINING TARGETS LIVE NEAR ZERO.

Central 98% shown · full tails stayed in the experiment · display scale is 100 × log growth

BIG ERRORS GET **SQUARED**. SO THEY CAN PULL THE MODELS AWAY FROM 0%.

A miss 5× as large gets 25× the squared-error penalty.

FIXED BENCHMARK

0% BASELINE

$$\hat{y} = 0$$

Always predicts 0%.
It never adjusts to reduce a miss.

ILLUSTRATIVE TRAINING ERRORS

2-unit miss $\rightarrow 2^2$

10-unit miss $\rightarrow 10^2$

4

VS

100

BIG MISSES COST MUCH MORE.

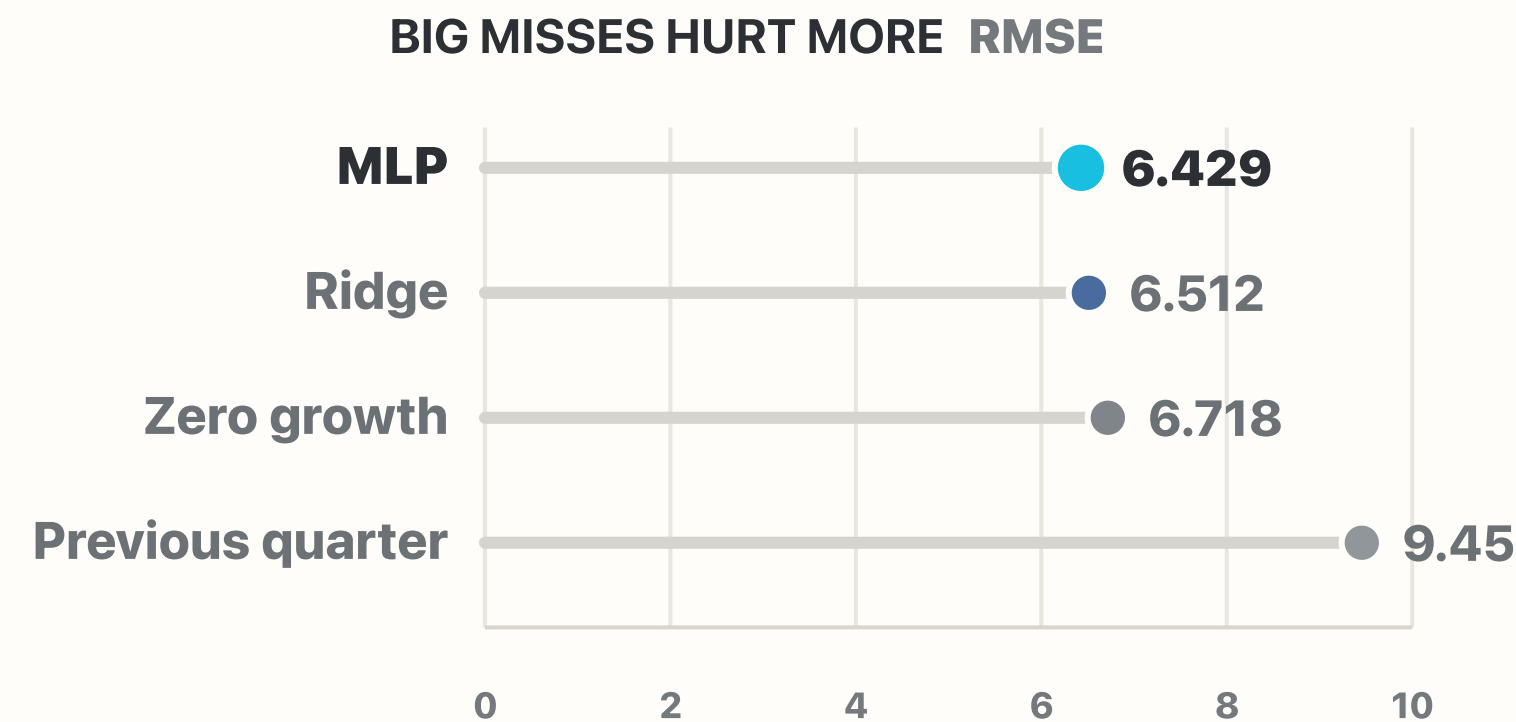
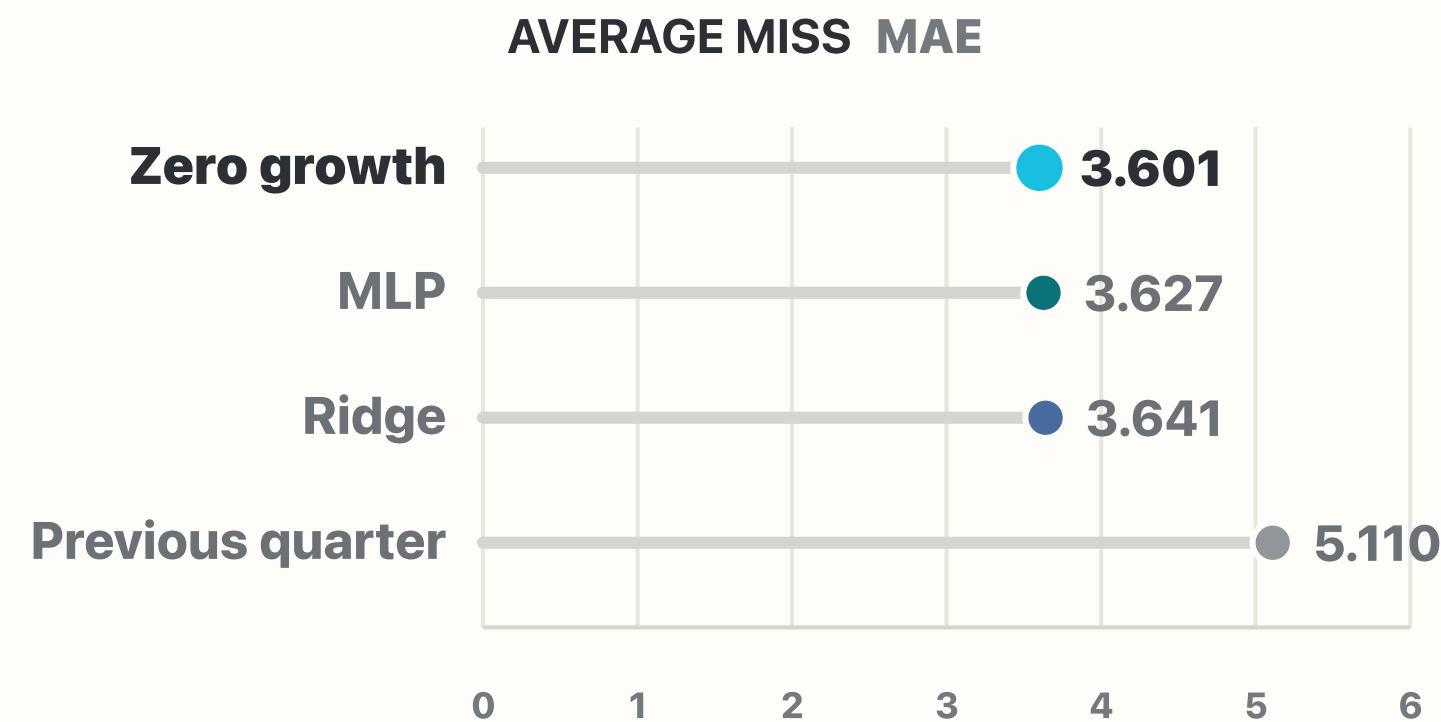
LEARNED MODELS

RIDGE + NEURAL NET

Can move away from 0% to reduce
those expensive large training misses.

**A LOWER SQUARED ERROR CAN COME WITH
A SLIGHTLY HIGHER AVERAGE ABSOLUTE MISS.**

ZERO WON MAE. THE NEURAL NETWORK WON RMSE.



“WINNER” DEPENDS ON WHAT KIND OF ERROR HURTS.

Same 13,532 evaluation examples · three already-examined 2024 quarters

RIDGE'S 454-BANK REVIEW LIST INCLUDED 112 OF THE QUARTER'S 454 LOWEST-GROWTH BANKS.

0% ties every bank. Ridge orders the queue by predicted growth, lowest first.

FIRST 10 OF 454 · MARCH 2024 FORECASTS FOR JUNE		
RANK	BANK	PREDICTED GROWTH
01	YOUNG AMERICANS BANK	-4.02%
02	PIONEER COMMERCIAL BANK	-3.32%
03	SUMITOMO MITSUI TR BK U SA	-2.83%
04	OLD GLORY BANK	-2.75%
05	TURTLE MOUNTAIN STATE BANK	-2.11%
06	COLUMBUS STATE BANK	-2.05%
07	FDS BANK	-2.03%
08	BOELUS STATE BANK	-1.96%
09	LEMONT NATIONAL BANK	-1.80%
10	CITIZENS NB OF CROSBYTON	-1.75%

LOWEST-GROWTH BANKS
CAPTURED
SAME 454 REVIEW SLOTS

Random selection

45.4 expected

Ridge

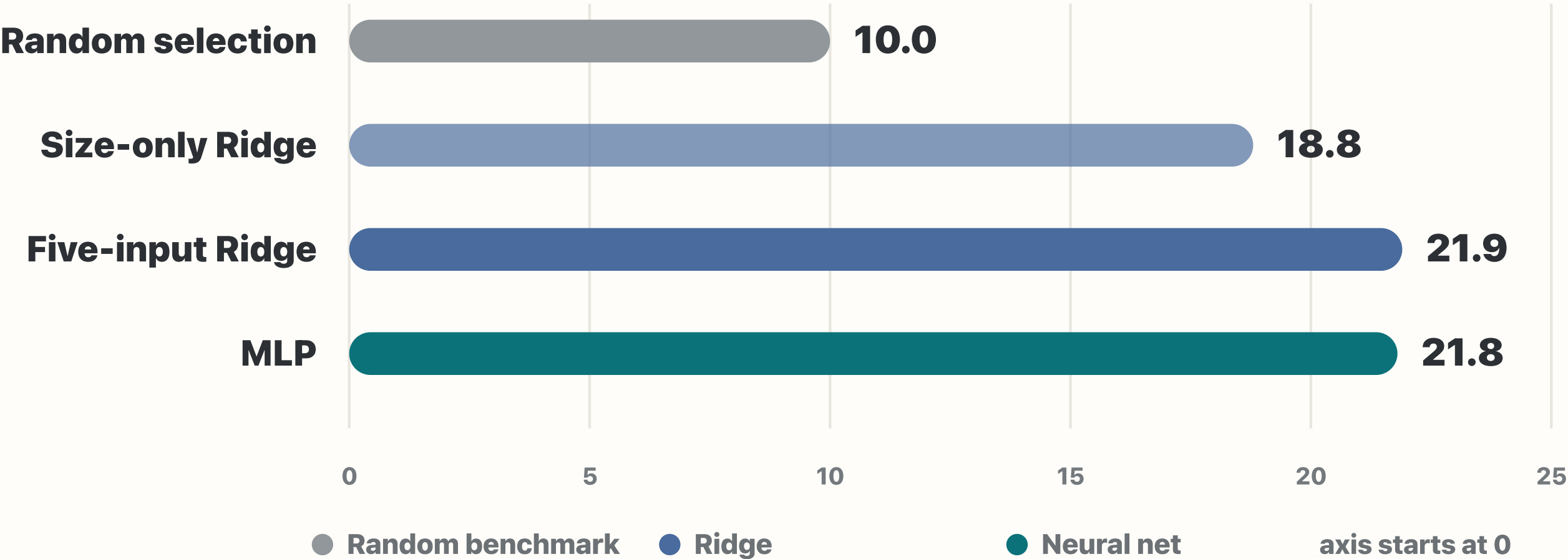
112 found

Neural network

109 found

4,536 eligible banks · March → June 2024. “Lowest-growth” = later in that quarter’s bottom 10% of deposit growth. It does not mean the bank was distressed. The 112 matches are across the full 454-bank queue, not just these ten rows.

I THREW OUT 4 INPUTS. KEPT ONLY DEPOSIT SIZE.



banks later in the lowest-growth 10% per 100 selected

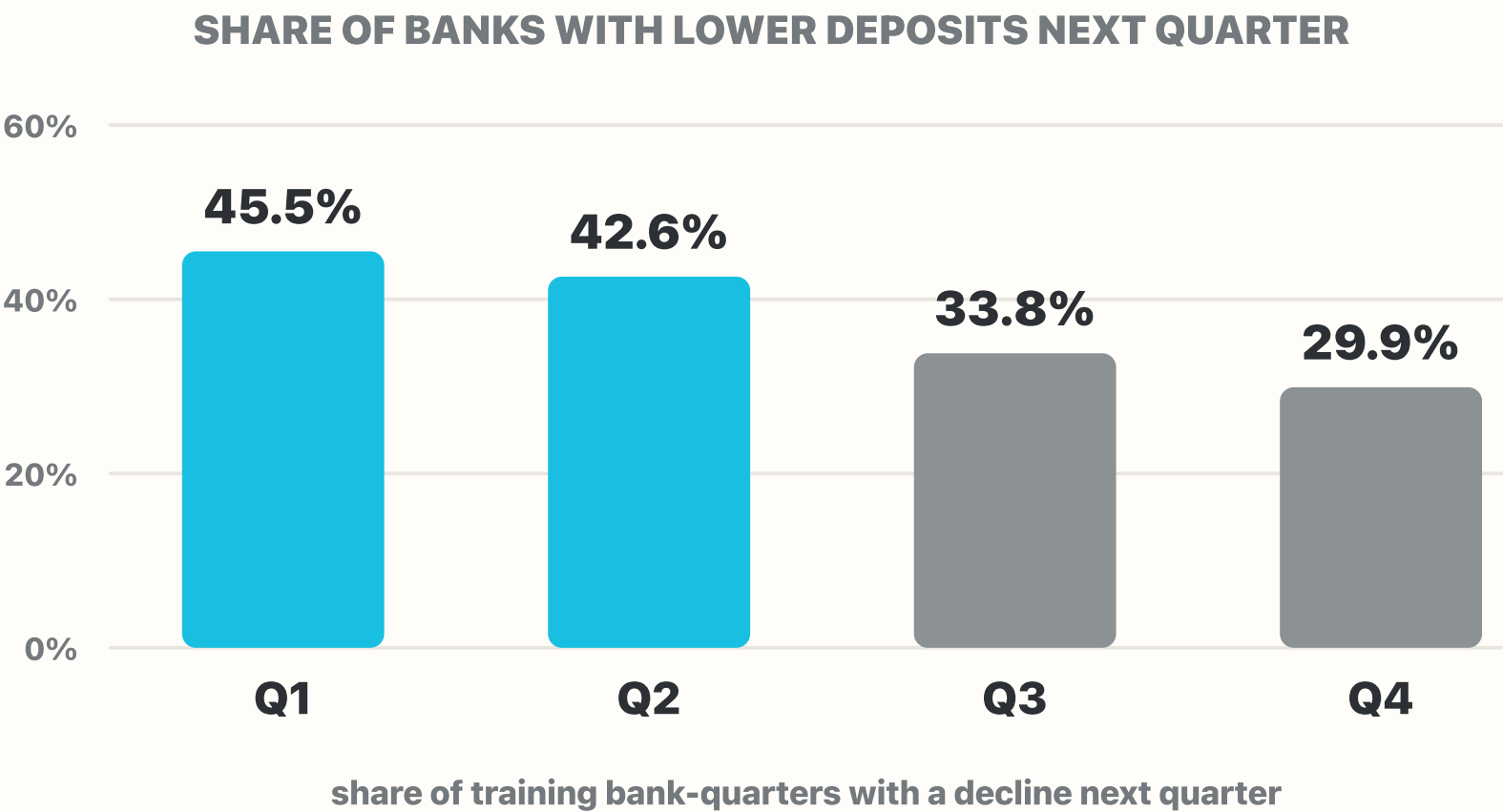
ONE INPUT GOT SURPRISINGLY CLOSE.

Equal-quarter averages across three reused 2024 quarters · size-only challenger added after examining 2024 · comparison is not a causal decomposition

DEPOSIT DECLINES WERE MORE COMMON AFTER Q1 THAN Q4.

Q1: 45.5% of training bank-quarters were followed by a deposit decline. **Q4: 29.9%.**

That is how often declines happened, not “banks lost 45.5% of their deposits.”



SO THE QUARTER ITSELF MIGHT BE USEFUL.

Training years only. This is a seasonal pattern in how often deposits declined next quarter.

SO I LET THE FORECAST **KNOW THE QUARTER.**

Instead of predicting 0% for everybody, use the **typical historical growth** for that calendar quarter.



HISTORICAL MAE · LOWER IS BETTER	
Seasonal rule	3.500
0% growth	3.601
IT WON ON REUSED 2024 DATA.	

NEXT UNSEEN QUARTER

DOES THE SEASONAL RULE **BEAT 0% AGAIN?**

Same banks. Same quarter. Lower MAE wins.

Caveat: I developed the seasonal rule after examining 2024. So this is a follow-up result that earns a fresh test, not a victory lap.